Novia One.

FMP · MODULE 5 · WORKBOOK

PRACTICAL EVIDENCE · INTEREST BEARING SECURITIES & BONDS

Workbook, answered.

Model answers to all **9 questions (50 marks)**, matched to the lecturer's marking style: bullet points with examples for list questions, prose for conceptual ones, and full workings shown for every calculation.

Note: study/model answers to learn from. The workbook is assessed as **your own work** (pass mark **65%**) — reword in your own voice for submission.

No lecturer memo has been released for Module 5 yet. When it is, the answers verified against it will carry a **MEMO** tag, exactly as in the Module 4 pack.

MARK ALLOCATION AT A GLANCE

QUESTION	TOPIC	MARKS
Q1	Treasury bills — definition, issuer and purpose of issuance	5
Q2	Main features of a bond · bond prices and interest rates	5
Q3	Under-priced (discount) versus premium bonds	3
Q4	Bond valuation at 4%, 6% and 7% — calculations	6
Q5	Duration versus investment horizon — which risk	2
Q6	Economic health and credit spreads	2
Q7	Effective duration from a valuation model — calculation	3
Q8	Case study — SA sovereign downgrade into junk (S&P, April 2020)	10
Q9	Multiple choice — valuation, quotes, embedded options, BEY, term structure	14
Total		**50**

ANSWER STYLE List questions use a lead-in plus **bold-label bullets** (with examples); conceptual and discursive questions are answered in **prose**, usually with a “However…” or consequence paragraph. Calculations show the **formula, the substitution and the answer**, plus the financial-calculator route.

ON THE MARK WEIGHTS The workbook gives marks **per question only**, not per sub-question. The weights shown in brackets on each card are therefore my **suggested split** — they add to the stated question total, but do not treat them as the lecturer's own allocation.

Before you submit

Name, surname and **ID number at the top of every page**, pages numbered · **Arial 12**, one-and-a-half line spacing, **2 cm margins** all round · number your answers to match the question numbers · complete the candidate details and declaration · submit as a **PDF** on Moodle by the deadline. Minimum **65%** to be found competent.

QUESTION 1 · TREASURY BILLS [05]

1.1 **Define a treasury bill and then indicate what institution issues it.** (≈3)

A **Treasury bill (T-bill)** is a **short-term debt instrument**, denominated in **South African rand**, that is **sold at a discount to its par value** and **carries no coupon**. Because it pays no periodic interest it is in effect a **zero-coupon instrument**: the investor's return is the difference between the discounted price paid and the par value, which is **redeemed in full at par on maturity**.

T-bills are issued at maturities ranging from **one day to twelve months**, with a typical life to maturity of **four weeks to twelve months**, which places them firmly in the **money market**. They carry **little or even no default risk**, and for that reason the interest rates on Treasury bills serve as the **benchmark default-free interest rates** in the market. Any new issue carrying the same maturity date as an existing issue is regarded as a **new tranche** of that security.

They are issued by **government (the National Treasury)**, and the **issuing agent is the South African Reserve Bank (SARB)**.

1.2 **For what purposes does this institution issue treasury bills?** (≈2)

Treasury bills are issued for the following purposes:

- **Short-term funding for government:** they raise short-term money to bridge the timing mismatch between when government collects revenue and when it must spend, and they help finance the **public sector borrowing requirement**.
- **Managing government's liquidity:** T-bills are used as a **tool to manage government's cash flow and liquidity position**, which is the money market's most important economic role — allowing economic units to adjust their liquidity positions.
- **Implementing monetary policy:** the SARB uses the money market, and the issue and redemption of government paper within it, to **manage the amount of money in circulation** and to transmit monetary policy decisions to the financial markets.
- **Providing a risk-free benchmark:** because they are effectively free of default risk, T-bill rates give the market its **benchmark default-free interest rates** against which other instruments are priced.
- **Providing a low-risk, highly liquid investment:** they give banks, corporates and money market funds a safe, marketable place to hold short-term surplus cash.

QUESTION 2 · FEATURES OF A BOND & THE PRICE–RATE RELATIONSHIP [05]

2.1 List and briefly describe the main features of a bond. (≈3)

The main features of a bond are:

- **Name of the issuer:** the entity borrowing the money and responsible for paying the interest and repaying the principal — governments, corporations or state-owned entities such as Eskom and Transnet.
- **Par (face) value:** the **principal amount repaid at maturity**, against which the coupon is calculated. Prices are quoted as a percentage of par, so a R1,000 par bond quoted at 98 sells for R980.
- **Coupon rate:** the rate applied to **par value** (not to the price) to determine the interest paid — a R1,000 bond with a 6% coupon pays R60 a year. It may be **fixed, floating or zero**.
- **Coupon frequency:** how often the coupon is paid — **usually semi-annually**; each payment is the annual coupon divided by the frequency.
- **Maturity date:** the date the principal falls due and the bond **ceases to exist**. Terms run from one day to 30 years (1–5 short, 5–12 medium, 12+ long); a bond with **no maturity date is a perpetual bond**.
- **Yield to maturity:** the **discount rate** used to price the bond — the effective return earned if the bond is held to maturity and the coupons are reinvested at that yield.
- **Currency:** the currency in which the coupon and principal are paid.
- **Security / collateral:** whether the bond is **secured** (backed by a legal claim on specified assets) or **unsecured** (a debenture, backed only by the issuer's promise to pay).
- **Contractual rights and covenants:** set out in the **trust deed** (the bond indenture) — the **negative covenants** that prohibit the borrower from certain actions and the **affirmative covenants** it promises to perform.

2.2 Describe your understanding of the relationship between bond prices and interest rates. (≈2)

Bond prices and interest rates move in **opposite directions** — they are **inversely related**. Once a bond has been issued its **coupon is fixed** for the life of the bond, but market interest rates change continuously, so newly issued bonds will offer different interest payments to existing ones.

Suppose interest rates **fall**. New bonds are issued with lower coupons, which makes the existing higher-coupon bonds **more valuable** to investors, so their **price rises**. Because the bond now costs more to buy, the return the new investor earns on it is lower — so the **yield falls**. The reverse happens when rates **rise**: new issues offer higher coupons, older low-coupon bonds become less attractive and must be **sold at a discount**, so their price falls and their yield rises.

Mathematically the same point holds because the **yield to maturity is the discount rate** in the pricing model: as the discount rate increases, the present value of the bond's fixed cash flows — and therefore its price — decreases. This is why the comparison of coupon against yield determines whether a bond trades at a **premium, at par or at a discount**, and it is the source of **interest rate risk** for anyone who may have to sell before maturity.

QUESTION 3 · UNDER-PRICED (DISCOUNT) VERSUS PREMIUM BONDS [03]

3.1 **Explain the difference between a bond that is under-priced and one that is trading at a premium.** (3)

The difference comes down to one comparison: the bond's **coupon rate against the prevailing market yield** (the required rate of return), which then fixes the price relative to **par**.

An **under-priced bond trades at a discount** — its price is **below par** (below 100). This happens when the bond's **coupon rate is lower than the yield the market currently requires**, typically because interest rates have risen since the bond was issued, or because the issuer's credit quality has deteriorated. An investor will only buy that older, lower-coupon bond if the price is low enough that the coupon plus the capital gain to maturity together deliver the market yield. The bondholder therefore earns part of the return as **capital appreciation**, because the price is **pulled up to par** as maturity approaches.

A bond **trading at a premium** is the mirror image — its price is **above par** (above 100), because its **coupon rate is higher than the yield the market currently requires**, typically because rates have fallen since issue or the issuer's credit standing has improved. Investors are willing to pay more than par for the above-market income stream, but the premium is steadily **amortised as the price declines back to par** at maturity, so the higher coupon is offset by a capital loss.

In both cases the bond **converges to par at maturity** — the constant-yield price trajectory — so the only difference is whether the investor is compensated through a capital gain (discount) or gives some of the higher coupon back through a capital loss (premium).

	DISCOUNT (UNDER-PRICED)	PREMIUM
Coupon vs yield	Coupon **<** market yield	Coupon **>** market yield
Price vs par	**Below** par (< 100)	**Above** par (> 100)
Usually because	Rates have **risen** / credit weakened	Rates have **fallen** / credit improved
To maturity	Capital **gain** as price rises to par	Capital **loss** as price falls to par

QUESTION 4 · BOND VALUATION AT THREE DISCOUNT RATES [06]

GIVEN A **R1,000 par**, **5-year**, **6% coupon** bond with **annual** interest payments. So **PMT = R60 a year** (6% × R1,000), **FV = R1,000** and **n = 5**. Required: the value of the bond when the discount rate is 4%, 6% and 7%.

Method — the present value model

The value of a bond is the **present value of its expected cash flows**, discounted at the required rate of return: the coupons form an annuity and the par value is a single lump sum at maturity.

$$P = \text{coupon} \times \frac{1 - (1 + i)^{-n}}{i} + \text{par} \times (1 + i)^{-n}$$

On a financial calculator: **N = 5, PMT = 60, FV = 1 000, I/Y = the discount rate → solve for PV**. (Interest is annual here, so there is no need to halve the coupon and rate or double the periods.)

4.1 Discount rate = 4% (2)

Annuity factor: $[1 - (1.04)^{-5}] \div 0.04$	**4.451822**
PV of coupons: R60 × 4.451822	**R267.11**
Discount factor: $(1.04)^{-5}$	**0.821927**
PV of par: R1,000 × 0.821927	**R821.93**
Value of the bond	**R1,089.04**

ANSWER **R1,089.04** — the bond trades at a **premium**, because the 6% coupon is higher than the 4% yield the market requires.

QUESTION 4, CONTINUED

4.2 Discount rate = 6% (2)

Annuity factor: $[1 - (1.06)^{-5}] \div 0.06$	**4.212364**
PV of coupons: R60 × 4.212364	**R252.74**
Discount factor: $(1.06)^{-5}$	**0.747258**
PV of par: R1,000 × 0.747258	**R747.26**
Value of the bond	**R1,000.00**

ANSWER **R1,000.00** — the bond trades **at par**, because the coupon rate and the required yield are **equal**.

4.3 Discount rate = 7% (2)

Annuity factor: $[1 - (1.07)^{-5}] \div 0.07$	**4.100197**
PV of coupons: R60 × 4.100197	**R246.01**
Discount factor: $(1.07)^{-5}$	**0.712986**
PV of par: R1,000 × 0.712986	**R712.99**
Value of the bond	**R959.00**

ANSWER **R959.00** — the bond trades at a **discount**, because the 6% coupon is lower than the 7% yield the market requires.

QUESTION 4 — THE THREE ANSWERS TOGETHER

DISCOUNT RATE (YTM)	VALUE OF THE BOND	TRADES AT
4% — below the coupon	**R1,089.04**	**Premium** (above par)
6% — equal to the coupon	**R1,000.00**	**Par**
7% — above the coupon	**R959.00**	**Discount** (below par)

WHAT THE THREE ANSWERS SHOW As the required rate of return rises from 4% to 7%, the value of exactly the same bond falls from R1,089.04 to R959.00 — the **inverse relationship between price and yield** in numbers, and the premium / par / discount rule: **coupon > yield → premium · coupon = yield → par · coupon < yield → discount**. Show the **formula, the substitution and the answer** for each part and **label it** premium, par or discount — the comparison earns the interpretation mark that the bare number does not.

The financial-calculator route

Enter **N = 5, PMT = 60, FV = 1 000**, then set **I/Y** to each rate in turn and **compute PV**:

I/Y = 4 → CPT PV	**−1,089.04**
I/Y = 6 → CPT PV	**−1,000.00**
I/Y = 7 → CPT PV	**−959.00**

The PV comes back **negative** because it is the cash you pay out — quote it as a positive value in your answer.

COMMON SLIPS ON THIS QUESTION Applying the coupon rate to the **price instead of par** (the coupon is always **6% × R1,000 = R60**, whatever the discount rate) · **halving the coupon and the rate** — this bond pays **annually**, so no semi-annual adjustment is needed · discounting the par value over the **wrong number of periods** (it is a single lump sum at **n = 5**, not an annuity).

QUESTION 5 · DURATION VERSUS INVESTMENT HORIZON [02]

5.1 **An investor has bought a bond with a 10-year duration but plans to sell it to buy a house in 5 years. What risk is he / she partaking in?** (2)

The investor is taking on **interest rate risk** — the risk that the price of the bond will fall following a rise in prevailing interest rates, leaving them with a **capital loss** when they sell.

The reason it is so acute here is the **mismatch between the bond's duration and the investor's holding horizon**. Duration measures how sensitive the bond's price is to a change in interest rates, and this bond has a duration of **10 years against a 5-year horizon**. As a rule of thumb the price moves by about 1% for every year of duration per 1% change in rates, so a **100 basis point rise in rates would cut the price by roughly 10%**. Because the bond will be **sold in the secondary market and not held to maturity**, the investor is **not guaranteed to receive par** — the proceeds will be whatever the market will pay at that time, which may be more or less than face value.

The way to manage it is to **manage duration**: matching the duration of the bond to the **five-year investment horizon** would largely immunise the investor against this risk.

QUESTION 6 · ECONOMIC HEALTH AND CREDIT SPREADS [02]

6.1 **Describe the relationship between the economic health of a nation and credit spreads.** (2)

The relationship is **inverse**: as the economic health of a nation **deteriorates, credit spreads widen**, and as the economy **strengthens, credit spreads narrow**.

A credit spread is the **difference in yield between a government (risk-free) bond and another debt security of the same maturity but lower credit quality** — it is the additional premium investors demand for taking on credit risk. When the economy weakens, company earnings and cash flows come under pressure, the **probability of default rises**, and investors' risk appetite falls, so they require **more compensation** to hold risky paper: the spread widens and, because yields and prices move inversely, bond prices fall. During an **economic expansion** the opposite happens — default risk recedes, demand for higher-yielding credit increases, and spreads narrow.

Because spreads move **systematically with the economy** in this way, they are widely watched as a **barometer of economic health**. The same logic applies at country level: the deeper a sovereign's credit standing falls, the wider the spread demanded over benchmark Treasuries — which is exactly what South Africa's downgrades into junk status produced.

QUESTION 7 · EFFECTIVE DURATION FROM A VALUATION MODEL [03]

GIVEN Current price P_0 **= R82.00** · if rates **decline** by 30 bp the price rises to P_- **= R83.50** · if rates **increase** by 30 bp the price falls to P_+ **= R80.75** · change in yield **Δy = 30 basis points = 0.0030**.

7.1 What is the duration of this bond? (3)

The portfolio manager is estimating **effective duration** — a direct measure of interest rate sensitivity taken from the prices the valuation model produces when the yield curve is shifted up and down:

$$\text{Effective duration} = \frac{P_- - P_+}{2 \times P_0 \times \Delta y}$$

Numerator: R83.50 − R80.75	**2.75**
Denominator: 2 × R82.00 × 0.0030	**0.492**
Effective duration: 2.75 ÷ 0.492	**5.59**

ANSWER The duration of the bond is approximately **5.59** (about **5.6 years**).

Interpretation. A duration of 5.59 means that for a **100 basis point (1%) change in interest rates** the bond's price will change by roughly **5.59% in the opposite direction** — so a 1% rise in rates implies a price fall of about 5.6%, and a 1% fall in rates implies a price rise of about 5.6%. The higher the duration, the more interest rate risk is embedded in the security.

TWO EASY MARKS TO LOSE First, **Δy must be entered as a decimal** — 30 basis points is **0.0030**, not 30 and not 0.30; using 0.30 gives 0.0559 and using 30 gives 559. Second, **P_- is the price when rates DECLINE** (the higher price, R83.50) and **P_+ the price when rates rise** (R80.75) — swapping them turns the answer negative.

Why duration and not convexity

Duration is a **linear** approximation of the price–yield relationship, so it is the right tool for a small ±30 bp shift. **Convexity** captures the curvature that duration misses, and it explains why duration slightly **under-estimates the price rise when yields fall** and **over-estimates the price fall when yields rise**.

QUESTION 8 · CASE STUDY — SA DOWNGRADED FURTHER INTO JUNK [10]

GIVEN On **29 April 2020 S&P Global Ratings** lowered South Africa's sovereign credit rating further into non-investment grade (junk), citing the impact of **COVID-19** on public finances and growth. The **long-term foreign currency** rating went from **BB to BB–** (three notches below investment grade) and the **long-term local currency** rating from **BB+ to BB** (two notches below). **Fitch** had downgraded weeks earlier for the lack of "a clear path towards government debt stabilisation", preceded by **Moody's** in March 2020. S&P expects the cost of servicing public debt to reach about **6.5% of GDP by 2023** and GDP to shrink **4.5%** (SARB forecast: 6.1%). The outlook was revised from **negative to stable**. The alert notes a **direct correlation between long-term interest rates and the depth of junk status** — the deeper the junk rating, the higher long-term rates tend to go.

8.1 Explain what credit rating is. (≈2)

A **credit rating** is an **assessment of the creditworthiness of the issuer** of a debt instrument — in effect a **prediction of the likelihood that the issuer will default** on its debt obligations. It is produced by a credit rating agency, which analyses both the **qualitative and quantitative** information about the prospective debtor, including non-public information obtained by its analysts, and then publishes the result on a standard scale running from **AAA (or Aaa)** for the highest credit quality down to **D or C** for default or "junk" quality.

Ratings matter because they drive the **return the issuer must offer**: a **lower rating requires a higher yield** to compensate investors for the additional risk they carry, and they also affect the instrument's **marketability**. A rating can be assigned to a **specific obligation** or to the **general creditworthiness of the entity** — the latter is a **sovereign credit rating**, which signifies a country's overall ability to provide a secure investment environment and takes account of its economic status, the transparency of its capital markets, investment flows, foreign direct investment, foreign currency reserves and political stability.

However, ratings are **not buy, sell or hold recommendations**. They measure only the issuing entity's **ability and willingness to repay debt**, they can be unreliable or inaccurate, and they should be used as only **one part of a credit analysis** alongside in-house research.

8.2 List any two (2) credit rating agencies. (≈2)

- **Standard & Poor's Global Ratings (S&P)**
- **Moody's Investors Service**

The third of the main agencies is **Fitch Ratings** — all three are named in the case study — and the study guide also lists **Morningstar DBRS**.

8.3 **Explain what a "downgrade" implies and how it affects both bond investors and issuers.** (≈3)

A **downgrade** is a rating agency's decision to **lower an issuer's credit rating**, and it implies a **deterioration in credit quality** — the agency now judges the probability of default to be higher than before. Where the downgrade takes the issuer **below investment grade**, as it did for South Africa, the debt is classified as **non-investment grade, speculative or "junk"**. This is the **downgrade risk** component of credit risk: an anticipated downgrade **increases the credit spread**, which in turn **decreases the price of the bonds**.

Effect on bond investors

- **Capital losses on existing holdings:** the required yield rises, so the market value of bonds already held **falls**.
- **Forced selling and shrinking demand:** a downgrade out of investment grade often makes the bond **ineligible for portfolios mandated to hold only investment-grade paper** — insurers, banks and pension funds that confine themselves to the top rating categories, and index-tracking funds — so they must sell into a falling market.
- **Higher return on new money:** investors who buy after the downgrade are **compensated with a higher yield**, which is why junk bonds are also called high-yield bonds; the case study notes investors will demand a higher rate of interest for lending.
- **Wider spreads and more volatility:** the spread over the risk-free benchmark widens, and for foreign investors a weaker currency can compound the loss.

Effect on issuers

- **Higher cost of borrowing:** the issuer must pay a higher rate of interest to raise the same money — the alert makes the point that the **further South Africa falls into junk status, the more long-term interest rates tend to rise**.
- **A heavier debt-service burden:** for the sovereign, S&P expects the cost of servicing public debt to climb to about **6.5% of GDP by 2023**, which consumes revenue that could have funded services.
- **Knock-on effect to other issuers:** the sovereign rating effectively sets a ceiling for the country's banks, state-owned entities and corporates, so **their** long-term unsecured debt is repriced too — which is why clients are told to check the rating provisions in their facilities.
- **Weaker demand and refinancing risk:** a smaller pool of eligible buyers may force the issuer to offer larger discounts, shorter maturities or to postpone issuance altogether.

8.4 **Describe how the real sector of the South African economy may be influenced by such a downgrade.** (≈3)

The **real sector** is the part of the economy that produces goods and services — firms investing, households consuming and workers earning — as opposed to the financial sector. A sovereign downgrade reaches it through the **cost and availability of money**.

- **Higher borrowing costs across the economy:** because long-term interest rates rise with the depth of junk status, and the government bond yield is the benchmark off which all other domestic debt is priced, **firms and households pay more to borrow**. Higher rates discourage **capital investment, expansion and job creation**, and they raise the cost of mortgages, vehicle finance and credit for consumers, which **reduces disposable income and consumption**.
- **Crowding out of government spending:** with debt service heading for about **6.5% of GDP**, a growing share of tax revenue goes to interest rather than to **infrastructure, health, education and social spending** — the very expenditure that supports real economic activity.
- **Reduced foreign investment:** the sovereign rating is the **first metric most institutional investors look at before investing internationally**, and many funds may not hold sub-investment-grade paper. Lower **portfolio and direct investment inflows** mean less capital for factories, plant and projects.
- **Capital outflows, a weaker rand and imported inflation:** as investors withdraw, the currency weakens, which raises the price of imported fuel, machinery and inputs. That feeds **inflation**, squeezes margins and household budgets, and may oblige the SARB to keep **policy rates higher** than growth would otherwise warrant.
- **Tighter credit and weaker banks' funding:** because the banks' own long-term debt is repriced after the sovereign downgrade, their funding costs rise and they **lend more cautiously** — hardest felt by small and medium enterprises.
- **Slower growth, lower confidence and employment losses:** in combination with COVID-19, S&P expected **GDP to shrink 4.5%** in that year. Weaker business and consumer confidence **defers investment and hiring**, which raises unemployment and, through lower tax revenue, feeds back into the fiscal position.

However, the picture is not uniformly negative: S&P revised the outlook from **negative to stable**, recognising some of government's fiscal and monetary measures as strong points, and National Treasury noted that **monetary policy was providing liquidity in the bond market and so reducing bond yields** — which cushions the real sector's borrowing costs in the short term.

QUESTION 9 · MULTIPLE CHOICE [14]

Seven questions at two marks each. Tick **one** box per question; the selected option is shown in green with the working underneath.

9.1 **A bond with a 5.5% coupon paid annually, maturing in three years, required rate of return 5%. The price per 100 of par is closest to:** (2)

A 98.65

B ✓ 101.36

C 106.43

Annuity factor: $[1 - (1.05)^{-3}] \div 0.05$	**2.723248**
PV of coupons: 5.5×2.723248	**14.98**
PV of par: $100 \times (1.05)^{-3} = 100 \times 0.863838$	**86.38**
Price per 100 of par	**101.36**

B · 101.36 The coupon (5.5%) is **above** the required return (5%), so the bond must trade at a **premium** — which rules out A immediately.

9.2 **Bond A (5% coupon, 2 years) and Bond B (3% coupon, 2 years), annual interest, market discount rate 4%. The price difference per 100 of par is closest to:** (2)

A 3.70

B ✓ 3.77

C 4.00

Annuity factor: $[1 - (1.04)^{-2}] \div 0.04$	**1.886095**
Bond A: $5 \times 1.886095 + 100 \times 0.924556$	**101.886**
Bond B: $3 \times 1.886095 + 100 \times 0.924556$	**98.114**
Difference: 101.886 − 98.114	**3.772**

B · 3.77 Both bonds share the same maturity and discount rate, so the difference is simply the **2% coupon differential discounted as a two-year annuity**: 2×1.886095 = **3.77**. Option C (4.00) is the trap — the undiscounted total of the two extra coupons.

9.3 **Bond A (101.886, 5%, 2 yrs), Bond B (100.000, 6%, 2 yrs), Bond C (97.327, 5%, 3 yrs). Which bond offers the lowest yield-to-maturity?** (2)

A ✓ Bond A

B Bond B

C Bond C

Compare each price with par, then confirm by pricing at the candidate yield:

BOND	PRICE VS PAR	SO THE YTM MUST BE...	YTM
A	101.886 — **premium**	**below** the 5% coupon	5 × 1.886095 + 100 × 0.924556 = 101.886 → **4%**
B	100.000 — **at par**	**equal** to the coupon	**6%**
C	97.327 — **discount**	**above** the 5% coupon	5 × 2.673012 + 100 × 0.839619 = 97.327 → **6%**

A · BOND A Bond A is the only bond trading at a **premium**, so it is the only one whose yield is **below** its coupon rate — **4%**, against 6% for both B and C. You can answer this from the premium/par/discount rule alone, without calculating a single yield.

9.4 **Bond dealers most often quote the:** (2)

A ✓ Flat price

B Full price

C Full price plus accrued interest

A · FLAT PRICE The **flat price** is also called the **clean or quoted price** — the clue is in the name. It is the **full price less accrued interest**, and dealers quote it so that the quote does not jump around simply because interest is accruing day by day.

The **full price** (invoice or **dirty** price) is what the buyer actually pays at settlement: **accrued interest + clean price = full price**, with the accrued interest **paid to the seller** for the period from the last coupon date to the sale date. Option C is nonsense — the full price **already includes** accrued interest, so adding it again double-counts.

9.5 **All else equal, which ONE of the following types of bonds will offer the greatest YTM at issuance?** (2)

A Putable bond

B ✓ Callable bond

C Convertible bond

B · CALLABLE BOND The **call option belongs to the issuer**, so it is valuable to the **issuer** and disadvantageous to the bondholder — who also carries the **reinvestment risk** of having the bond redeemed exactly when proceeds can only be reinvested at a lower yield. The bondholder must therefore be compensated, so a callable bond must offer a **higher yield (a lower price)** than an otherwise identical non-callable bond.

The other two options give the **option to the bondholder**: a **putable** bond lets the holder sell it back at a prespecified price, and a **convertible** lets the holder exchange it for shares. Because those rights are valuable to the holder, both are issued at a **lower yield (a higher price)** than an identical straight bond.

9.6 **Calculate the BEY for a money market security with 180 days to maturity and a quoted add-on yield of 2% based on a 365-day year.** (2)

A 2.12%

B 2.63%

C ✓ 2.0%

Holding period yield: 2% × (180 ÷ 365)	**0.9863%**
BEY = HPY × (365 ÷ 180) = 0.9863% × 2.02778	**2.00%**

C · 2.0% The **bond equivalent yield is itself an add-on yield expressed on a 365-day year** — the nominal annual rate. The quote given is **already** an add-on yield on a 365-day basis, so no conversion is needed and the **BEY equals the quoted 2%**. Annualising the 180-day holding period yield returns the same 2.0%, as the working shows.

Where a conversion *is* required is when the quote is on a different basis — for example a **discount yield**, which conventionally uses a **360-day** year: 2% × 365/360 = 2.03%. This is the one item in the paper whose distractors do not reconcile with a standard route, so **check it against the memo** when it is released.

9.7 **Under the pure expectations theory, an inverted yield curve is interpreted as evidence that...** (2)

A demand for long-term bonds is falling

B investors have very little demand for liquidity

C ✓ short-term rates are expected to fall in the future

C · SHORT-TERM RATES ARE EXPECTED TO FALL Under **pure expectations**, any long-term rate is the **geometric mean of current and expected future short-term rates**, so the shape of the curve is read purely as an expectation. An **inverted** curve means **short-term rates are higher now but are expected to decrease in future**, which is also why the material links an inverted curve to expectations of a **lower future policy rate**.

A describes a **market segmentation / supply-and-demand** explanation and B a **liquidity preference** explanation — both are the wrong theory for this stem.

CALCULATION SUMMARY

CALCULATION	FORMULA USED	ANSWER
Q4.1 — value at 4%	PV of coupon annuity + PV of par	**R1,089.04** — premium
Q4.2 — value at 6%	PV of coupon annuity + PV of par	**R1,000.00** — at par
Q4.3 — value at 7%	PV of coupon annuity + PV of par	**R959.00** — discount
Q7 — duration	$(P_{-} - P_{+}) \div (2 \times P_0 \times \Delta y)$	**5.59** (2.75 ÷ 0.492)

MULTIPLE CHOICE — ANSWER KEY

QUESTION	ANSWER	WHY
9.1	**B** — 101.36	Coupon 5.5% > required 5% → premium; PV = 14.98 + 86.38
9.2	**B** — 3.77	2% coupon differential × two-year annuity factor 1.886095
9.3	**A** — Bond A	Only bond at a premium → YTM 4%, below B and C at 6%
9.4	**A** — Flat price	Flat = clean = quoted price; full price includes accrued interest
9.5	**B** — Callable bond	Option belongs to the **issuer** → higher yield / lower price
9.6	**C** — 2.0%	BEY is an add-on yield on a 365-day year — the quote already is one
9.7	**C** — short-term rates expected to fall	Pure expectations reads the curve as pure expectation of future rates

WHEN THE MEMO LANDS Send me the lecturer's answers and I will re-cut this pack against them — tagging each verified answer **MEMO** and adjusting any wording, mark split or answer that differs, exactly as we did for Module 4. The two items most worth checking are **9.6 (BEY)** and the **mark splits within Q1, Q2 and Q8**.

Study pair

Use this pack with the two Module 5 study documents: the **Summarised Reference** for last-minute recall and the **In-Depth Workbook** for understanding. All three are built from Novia's own study guide and class deck.